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Abstract

This project presents an automated system for detecting defects in solar cells using a combination of deep learning and machine learning techniques. The system utilizes electroluminescence (EL) images from the ELPV and EL datasets, which contain both normal and defective solar cells with varying defect levels. Multiple convolutional neural network (CNN) architectures, including ResNet, GoogLeNet, Xception, DarkNet, SqueezeNet, CNet, and Vision Transformer (ViT), are used as feature extractors to capture important visual patterns from the images. The extracted features are then classified using machine learning models such as Support Vector Machine (SVM) and K-Nearest Neighbors (KNN).

The performance of the models is evaluated using two different image resolutions (224×224 and 64×64) to analyze the trade-off between accuracy and computational efficiency. Experimental results show that the Xception model achieves the highest accuracy among all models, while reducing image size decreases computational cost but slightly affects performance. The study demonstrates that combining deep learning-based feature extraction with traditional machine learning classifiers provides an effective and scalable approach for solar panel defect detection. This system can help improve the efficiency and reliability of solar energy systems by enabling faster and more accurate defect identification.

The performance of the models is evaluated using two different image resolutions (224×224 and 64×64) to analyze the trade-off between accuracy and computational efficiency. Experimental results show that the Xception model achieves the highest accuracy among all models, while reducing image size decreases computational cost but slightly affects performance. The study demonstrates that combining deep learning-based feature extraction with traditional machine learning classifiers provides an effective and scalable approach for solar panel defect detection. This system can help improve the efficiency and reliability of solar energy systems by enabling faster and more accurate defect identification.

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Acronyms

- AI – Artificial Intelligence
- ML – Machine Learning
- DL – Deep Learning
- CNN – Convolutional Neural Network
- ViT – Vision Transformer
- EL – Electroluminescence
- ELPV – Electroluminescence Photovoltaic Dataset
- PV – Photovoltaic
- SVM – Support Vector Machine
- KNN – K-Nearest Neighbors
- RGB – Red Green Blue
- GPU – Graphics Processing Unit
- CPU – Central Processing Unit
- ReLU – Rectified Linear Unit
- FC – Fully Connected Layer

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Chapter 1 Introduction

Solar energy has become one of the most important renewable energy sources due to its sustainability and environmental benefits. Photovoltaic (PV) systems, commonly known as solar panels, play a key role in converting sunlight into electrical energy. However, the efficiency of these systems can be significantly affected by defects such as cracks, corrosion, and hotspots in solar cells. Detecting these defects at an early stage is essential to maintain performance and ensure long-term reliability.

Traditional methods of defect detection rely on manual inspection or basic image processing techniques, which are time-consuming and often inaccurate for large-scale applications. With advancements in artificial intelligence, particularly in deep learning and computer vision, automated systems can now analyze images more efficiently and accurately.

In this project, an automated defect detection system is developed using electroluminescence (EL) images. Deep learning models are used to extract meaningful features from the images, and machine learning classifiers such as Support Vector Machine (SVM) and K-Nearest Neighbors (KNN) are applied for classification. This approach aims to improve the accuracy and efficiency of solar panel defect detection.

1.1 Background

Solar energy is one of the fastest-growing sources of renewable energy due to its sustainability and low environmental impact. Photovoltaic (PV) systems, commonly known as solar panels, are widely used to convert sunlight into electrical energy. The performance and efficiency of these systems largely depend on the condition of the solar cells. However, defects such as cracks, corrosion, hotspots, and broken cells can occur during manufacturing, transportation, or long-term usage, leading to reduced energy output and potential system failure.

To identify these defects, electroluminescence (EL) imaging is commonly used, as it allows the visualization of internal defects that are not visible to the naked eye

Traditionally, defect detection has been performed manually or using basic image processing techniques, which are time-consuming, less accurate, and not suitable for large-scale industrial applications.

With the advancement of artificial intelligence, particularly in deep learning and computer vision, automated defect detection systems have gained significant attention. Convolutional Neural Networks (CNNs) are widely used for extracting meaningful features from images, while machine learning algorithms can be used for classification tasks. These approaches enable faster, more accurate, and scalable defect detection, making them highly suitable for modern solar energy systems.

1.2 Problem Statement

The efficiency and reliability of solar panels are significantly affected by defects such as cracks, corrosion, and other structural abnormalities in solar cells. Traditional defect detection methods, which rely on manual inspection or basic image processing techniques, are time-consuming, labor-intensive, and prone to human error. These methods are not suitable for large-scale solar installations where a large number of panels need to be inspected regularly.

Furthermore, detecting subtle and complex defects in electroluminescence (EL) images is challenging using conventional approaches, as they often fail to capture fine-grained features and variations in defect patterns.

This creates a need for an automated and efficient system that can accurately analyze solar cell images and detect defects with high reliability.

Therefore, the main problem addressed in this project is to develop an automated defect detection system using deep learning and machine learning techniques that can effectively identify and classify defects in solar cells while improving accuracy, reducing manual effort, and ensuring scalability for real-world applications.

1.3 Motivation

The increasing global demand for clean and renewable energy has made solar power a critical component of modern energy systems. Ensuring the efficiency and reliability of solar panels is essential for maximizing energy output and reducing operational costs. However, defects in solar cells can significantly degrade performance if not detected early.

Manual inspection methods are not only time-consuming but also unreliable when dealing with large-scale solar installations. This creates a strong need for an automated and accurate defect detection system. With recent advancements in artificial intelligence, particularly in deep learning and computer vision, it is now possible to analyze complex image data and detect subtle defects effectively.

1.4 Objectives

- To develop an automated system for detecting defects in solar cells using image data.
- To extract meaningful features from electroluminescence (EL) images using deep learning models.
- To use CNN architectures such as ResNet, GoogLeNet, Xception, DarkNet, SqueezeNet, CNet, and ViT for feature extraction.
- To apply SVM and KNN for defect classification.
- To compare model performance based on accuracy and efficiency.
- To analyze the effect of different image resolutions (224×224 and 64×64).
- To design a scalable and efficient defect detection system.

1.5 Scope of the Project

This project focuses on detecting defects in solar cells using electroluminescence (EL) images from datasets such as ELPV and EL. The work includes data preprocessing, feature extraction using different convolutional neural network (CNN) models, and classification using machine learning algorithms like Support Vector Machine (SVM) and K-Nearest Neighbors (KNN). The performance of the system is evaluated using different image resolutions to study the trade-off between accuracy and computational efficiency.

The scope is limited to image-based defect classification and does not include real-time deployment or hardware implementation.

1.6 Organization of the Report

- Chapter 1: Introduction – Covers background, problem statement, motivation, objectives, and scope of the project.
- Chapter 2: Literature Review – Discusses existing research and techniques related to solar panel defect detection.
- Chapter 3: Methodology – Explains dataset details, data preprocessing, feature extraction, model implementation, and evaluation metrics.
- Chapter 4: Results and Discussion – Presents experimental results, model performance, and analysis.
- Chapter 5: Conclusion and Future Work – Summarizes the findings and suggests future improvements.

Chapter 2 Literature Review

1. Solar panel defect detection has evolved from traditional manual inspection and basic image processing techniques to advanced automated methods using artificial intelligence, improving efficiency and scalability [4][5].
2. Conventional approaches are often time-consuming and less effective in detecting subtle defects, especially when dealing with large-scale datasets and complex defect patterns [4].
3. Electroluminescence (EL) imaging is widely used as a non-destructive technique for identifying internal defects such as cracks, corrosion, and microfractures in photovoltaic cells [2][3].
4. Recent research focuses on applying deep learning techniques, particularly Convolutional Neural Networks (CNNs), to automatically extract features from EL images and enhance detection accuracy [2][3][6].
5. Advanced CNN architectures such as ResNet, GoogLeNet, and Xception have demonstrated strong performance in defect detection tasks due to their ability to learn deep and complex features [1][3][6].
6. Self-supervised and hybrid deep learning frameworks are also being explored to reduce dependency on labeled data and further improve detection performance in real-world industrial applications [1].

2.1 Solar Panel Defect Detection

Solar panel defect detection is important for maintaining the efficiency and reliability of photovoltaic (PV) systems. Defects such as cracks, corrosion, hotspots, and broken cells can reduce power output and affect system performance.

Traditional detection methods rely on manual inspection and basic image processing, which are time-consuming and less accurate for large-scale applications. These methods also struggle to detect subtle defects.

With the advancement of artificial intelligence and computer vision, automated approaches have been developed. Deep learning techniques, in particular, can extract meaningful features from images and provide more accurate and efficient defect detection.

2.2 Electroluminescence (EL) Imaging

Electroluminescence (EL) imaging is a widely used technique for inspecting solar cells and detecting internal defects. It works by passing an electric current through the solar cell, which causes it to emit light. The emitted light is captured using specialized cameras to produce detailed images of the cell structure.

EL images are highly effective in revealing defects such as cracks, broken cells, corrosion, and inactive regions that are not visible through normal visual inspection. These images provide valuable information about the internal condition of solar cells, making them essential for quality control and maintenance.

Due to its ability to highlight subtle defects, EL imaging is commonly used in research and industry. It also serves as an important data source for machine learning and deep learning models used in automated defect detection systems.

2.3 Deep Learning for Defect Detection

Deep learning has significantly improved image-based defect detection by enabling automatic feature extraction from raw data. Convolutional Neural Networks (CNNs) are widely used in this domain due to their ability to learn complex patterns and identify subtle variations in images.

Models such as ResNet, GoogLeNet, and Xception have shown strong performance in detecting defects in solar cells by capturing both low-level and high-level features. These models reduce the need for manual feature engineering and provide higher accuracy compared to traditional methods. Recently, advanced architectures like Vision Transformer (ViT) have also been explored to capture global relationships within images.

2.4 Machine Learning Classifiers (SVM, KNN)

Machine learning classifiers play an important role in categorizing data based on extracted features. In this project, Support Vector Machine (SVM) and K-Nearest Neighbors (KNN) are used for defect classification.

SVM is a supervised learning algorithm that finds the optimal boundary (hyperplane) to separate different classes. It is effective for high-dimensional data and provides good generalization performance. On the other hand, KNN is a simple and intuitive algorithm that classifies a sample based on the majority class of its nearest neighbors. It works well when the data is properly structured and labeled.

When combined with deep learning-based feature extraction, these classifiers can achieve good performance by utilizing meaningful features while maintaining lower computational complexity compared to fully deep learning-based classification systems.

2.5 Summary of Existing Research

Recent studies in solar panel defect detection mainly use deep learning techniques, especially CNNs, to improve accuracy in analyzing EL images. Models like ResNet and Xception have shown better performance than traditional methods.

Machine learning algorithms such as SVM and KNN are often combined with deep learning features for efficient classification. Although advanced methods improve performance, they can be complex and require large datasets. Therefore, simpler approaches that balance accuracy and efficiency are still important.

Chapter 3 — Methodology

This chapter describes the overall approach used for solar panel defect detection. The methodology consists of data collection, preprocessing, feature extraction, classification, and evaluation of the models. Electroluminescence (EL) images from the ELPV and EL datasets are used as input for the system.

Initially, the dataset is organized and preprocessed, including resizing images and splitting them into training, validation, and testing sets. Different image resolutions (224×224 and 64×64) are used to analyze the impact on model performance After preprocessing, deep learning models such as ResNet, GoogLeNet, Xception, DarkNet, SqueezeNet, CNet, and Vision Transformer (ViT) are used to extract meaningful features from the images.

The extracted feature vectors are then passed to machine learning classifiers, specifically Support Vector Machine (SVM) and K-Nearest Neighbors (KNN), for defect classification. The performance of the models is evaluated using accuracy as the primary metric. This methodology provides an efficient pipeline that combines deep learning and machine learning techniques for accurate defect detection.

3.1 Dataset Description

- Two datasets are used: ELPV and EL for solar cell defect detection.
- Both datasets contain electroluminescence (EL) images that highlight internal defects.
- The ELPV dataset includes monocrystalline and polycrystalline solar cell images with defect probability labels (0%, 33%, 67%, 100%).
- The EL dataset contains images classified as normal, cracked, and corroded cells.
- The datasets are divided into training, validation, and testing sets.
- Images are grouped into functional, mild, moderate, and severe defect categories for analysis.

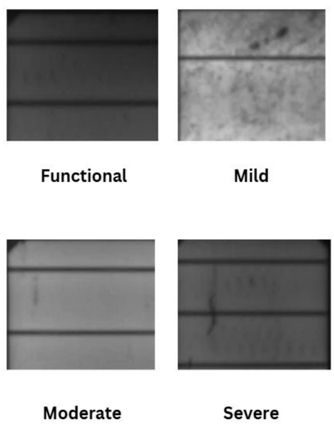

Fig 1.1: Sample EL Images[7]

Fig 1.2: Class Distribution[7]

3.2 Data Preprocessing

- The dataset is organized and cleaned to remove inconsistencies.
- Images are resized to fixed dimensions (224×224 and 64×64) for model input.
- The data is split into training, validation, and testing sets.
- Images are labeled and categorized into different defect levels.
- Normalization is applied to standardize pixel values.
- Data balancing techniques are used to handle class imbalance.

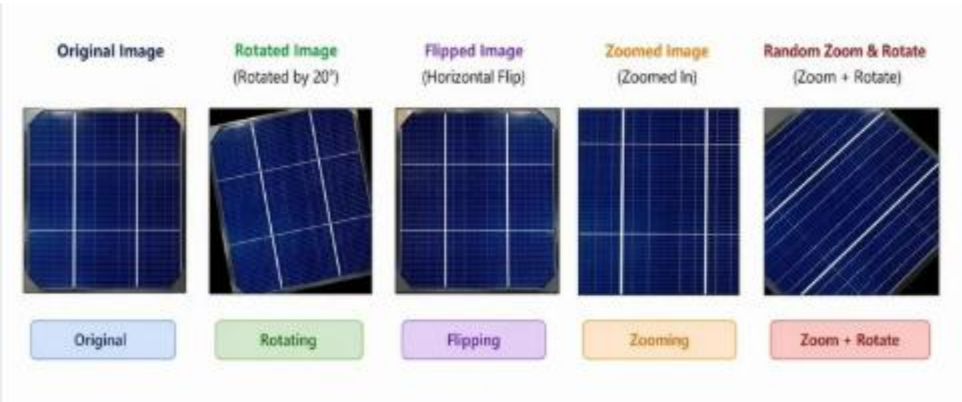

Fig 1.3: Augmented Images[8]

Table 1-1: Data Augmentation Parameters

Parameter	Value	Purpose
Rotation Range	±25°	Handle tilted cell images
Zoom Range	0.2	Achieve scale invariance
Width/Height Shift	0.1	Provide position invariance
Horizontal Flip	TRUE	Capture mirror symmetry
Fill Mode	nearest	Fill shifted pixels
Rescaling	1/255	Normalize pixel values [0,1]

3.3 System Architecture

- The proposed system architecture is designed to detect defects in solar cells using electroluminescence (EL) images. The input images first undergo preprocessing steps such as resizing and normalization to ensure consistency. These processed images are then passed through various convolutional neural network (CNN) models, including ResNet, GoogLeNet, Xception, DarkNet, SqueezeNet, CNet, and Vision Transformer (ViT), which extract meaningful features from the images.

- The extracted feature vectors are then provided to machine learning classifiers such as Support Vector Machine (SVM) and K-Nearest Neighbors (KNN). These classifiers analyze the features and predict whether the solar cell is defective or non-defective. This architecture effectively combines deep learning for feature extraction and machine learning for classification, defect detection system.

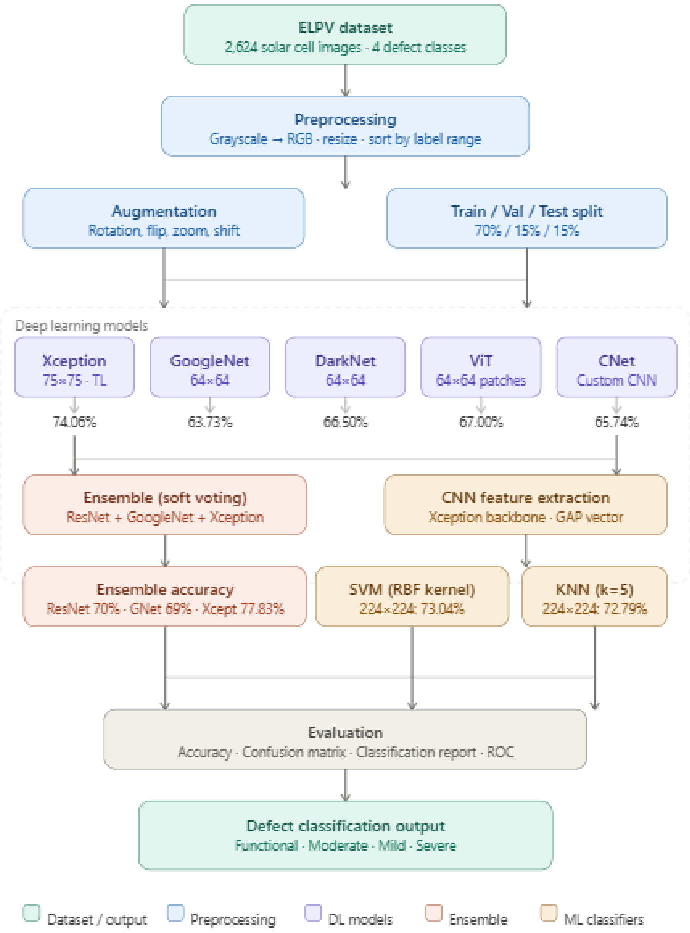

Fig1.4: System Architecture[1-6]

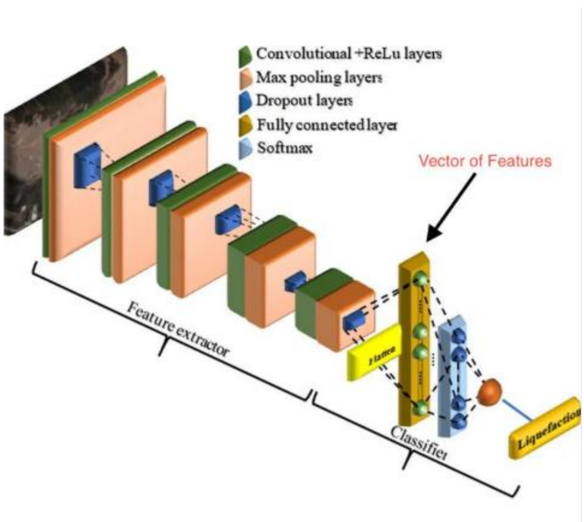

Fig 1.5: Proposed System Architecture[8]

3.4 Feature Extraction using CNN Models

- Deep learning models are used to extract features from EL images automatically.
- CNNs learn patterns such as edges, textures, and defect structures.
- Models used include ResNet, GoogLeNet, Xception, DarkNet, SqueezeNet, CNet, and ViT.
- Each model generates a feature vector representing important image information.
- These features are used as input for machine learning classifiers, reducing the need for manual feature extraction.

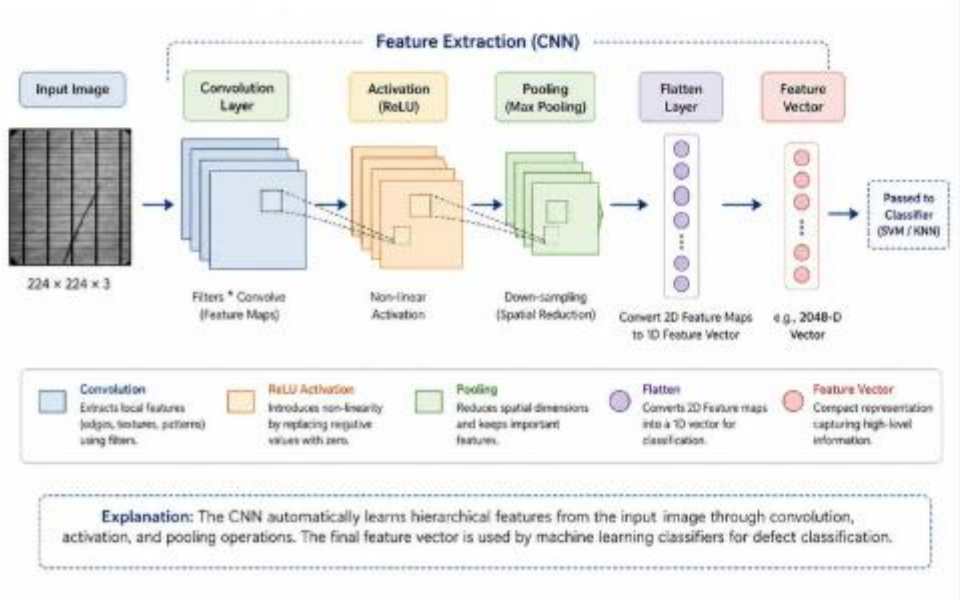

Fig 1.6: CNN Feature Extraction Process[8]

3.5 Machine Learning Models Used (SVM, KNN)

- Machine learning models are used for classifying defects based on extracted features.
- Support Vector Machine (SVM) is used to separate different classes by finding an optimal decision boundary.
- SVM performs well for high-dimensional feature data.
- K-Nearest Neighbors (KNN) classifies samples based on similarity with nearest data points.
- KNN is simple and effective for pattern-based classification.
- Both models use feature vectors obtained from CNN models.
- The final output is the predicted defect class of the solar cell.

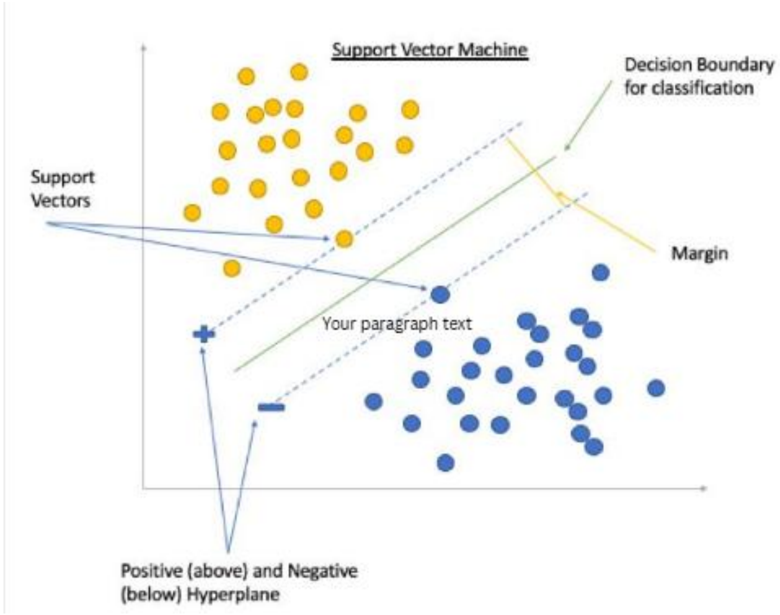

Fig 1.7: SVM Classification Visualization[8]

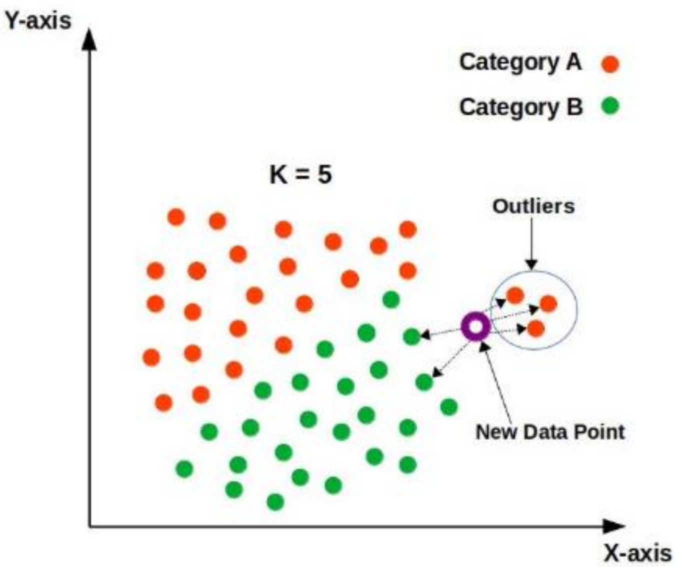

Fig 1.8: KNN Classification Visualization[8]

3.6 Model Training and Validation

- The dataset is divided into training, validation, and testing sets.
- CNN models are used to extract features from training images.
- The extracted features are used to train SVM and KNN classifiers.
- The validation set is used to evaluate model performance during training.
- Hyperparameters are adjusted to improve model accuracy.

- The final model is tested on unseen test data to measure performance.

3.7 Evaluation Metrics

- Model performance is evaluated using accuracy as the primary metric.
- Accuracy measures the ratio of correctly predicted samples to total samples.
- It helps in assessing how well the model classifies defects.
- Higher accuracy indicates better model performance.
- The metric is used to compare different models and configurations.

Chapter 4 — Results and Discussion

- This chapter presents the results obtained from the implemented models and analyzes their performance in detecting solar cell defects. Various deep learning models are evaluated for feature extraction, and machine learning classifiers such as SVM and KNN are used for classification.
- The performance of the models is compared using different image resolutions (224×224 and 64×64) to study the impact on accuracy and efficiency. The results highlight the effectiveness of different models, with particular emphasis on identifying the best-performing approach. This chapter also provides a comparative analysis and interpretation of the results to understand the strengths and limitations of the proposed system.

Table 2-1: Performance of CNN Models at 224×224 Resolution

Model	Accuracy (%)
ResNet18	70.00%
GoogleNet	69.02%
Xception	77.83%
DarkNet	66.25%
SqueezeNet	67.25%
CNet	69.02%

4.1 Performance of CNN Models (224×224)

- CNN models are evaluated using images of size 224×224.
- Models used include ResNet, GoogLeNet, Xception, DarkNet, SqueezeNet, and CNet.
- Xception achieved the highest accuracy among all models.
- ResNet and GoogLeNet showed moderate performance.
- DarkNet, SqueezeNet, and CNet performed comparatively lower.
- Higher image resolution helps capture detailed features, improving accuracy.
- Overall, Xception proved to be the most effective model at this resolution.

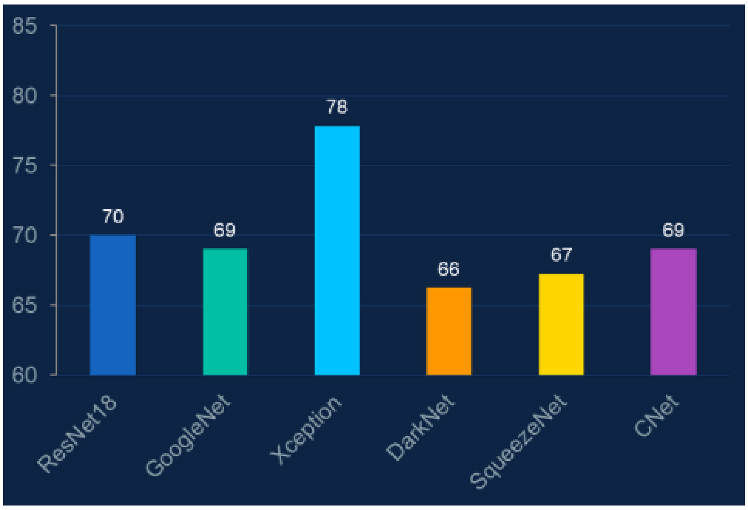

Fig 2.1: Model Performance Comparison (224x224)

4.2 Performance of CNN Models (64×64)

- CNN models are evaluated using images of size 64×64.
- Models used include GoogLeNet, CNet, DarkNet, Xception, and ViT.
- Xception achieved the highest accuracy among all models at this resolution.
- Other models showed reduced performance compared to higher resolution images.
- Lower image resolution reduces computational cost but also decreases accuracy.
- Important image details may be lost at smaller sizes, affecting model performance.
- Overall, performance is lower than 224×224, but faster and more efficient.

Table 2-2: Performance of CNN Models at 64×64 Resolution

Model	Accuracy (%)
GoogleNet	64%
CNet	66%
DarkNet	67%
Xception	74%
ViT	67%

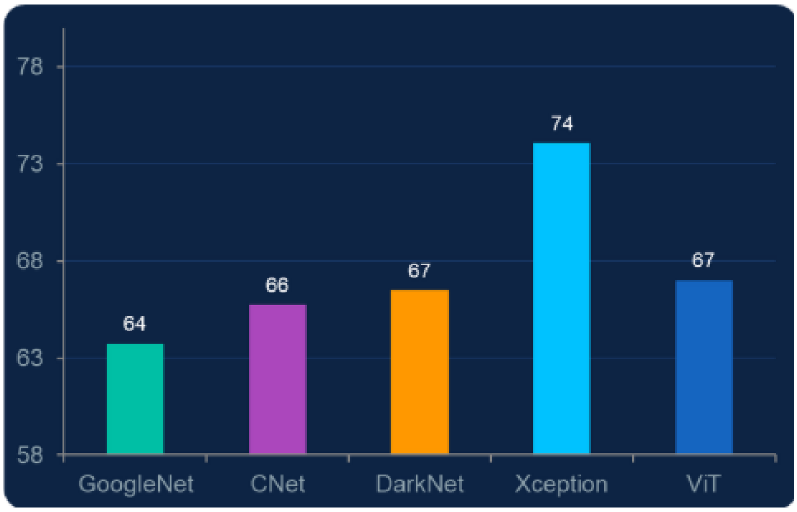

Fig 2.2: Model Performance Comparison (64x64)

Table 2-3: Models Used in Ensemble

Model	Accuracy (%)
ResNet	70.00%
GoogleNet	68.02%
Xception	77.83%

4.3 Performance of SVM and KNN

- SVM and KNN are used for classifying defects based on extracted features.
- SVM achieved slightly higher accuracy compared to KNN.
- Both models performed well when combined with CNN feature extraction.
- Performance improved with higher resolution images (224×224).
- KNN showed stable results but was slightly less accurate than SVM.
- The combination of CNN features with ML classifiers proved effective for defect detection.

4.4 Comparison of Model Performance

- Different CNN models are compared based on their accuracy and efficiency.
- Xception achieved the highest performance among all models.
- ResNet and GoogLeNet showed moderate results.
- DarkNet, SqueezeNet, and CNet performed comparatively lower.
- Higher image resolution (224×224) resulted in better accuracy than 64×64.
- SVM performed slightly better than KNN in classification tasks.
- Overall, the combination of Xception with SVM gave the best results.

Table 2-4: Effect of Image Size on SVM and KNN Performance

Model	Image Size	Accuracy (%)
SVM (RBF)	224×224	73.04%
SVM (RBF)	64×64	68.02%
KNN (k=5)	224×224	72.79%
KNN (k=5)	64×64	69.26%

4.5 Effect of Image Resolution

- Image resolution has a significant impact on model performance.
- Higher resolution (224×224) provides more detailed features, improving accuracy.
- Lower resolution (64×64) reduces computational cost and processing time.
- However, smaller images may lose important details, leading to lower accuracy.
- Models perform better with higher resolution but require more resources.
- A trade-off exists between accuracy and efficiency based on image size.

4.6 Interpretation of Results

- Xception performed the best due to its efficient feature extraction capability.

- Higher resolution images improved accuracy by capturing finer defect details.
- Lower resolution images reduced performance due to loss of important features.
- SVM showed slightly better results than KNN for classification.
- CNN + machine learning combination proved effective for defect detection.
- The system achieved a good balance between accuracy and computational efficiency.

Chapter 5 — Conclusion and Future Work

- This chapter summarizes the overall findings of the project and highlights the effectiveness of the proposed defect detection system. The results demonstrate that combining deep learning for feature extraction with machine learning classifiers provides accurate and efficient defect detection.
- The chapter also discusses the limitations of the current approach and suggests possible improvements for future work. These enhancements aim to further improve accuracy, scalability, and real-world applicability of the system.

5.1 Summary of Findings

- The proposed system successfully detects defects in solar cells using EL images.
- CNN models effectively extract meaningful features from images.
- Xception achieved the highest accuracy among all tested models.
- Higher image resolution (224×224) provided better performance than 64×64.
- SVM performed slightly better than KNN for classification.
- The combination of deep learning and machine learning proved efficient and accurate.

5.2 Limitations

- The model performance depends on the quality and size of the dataset.
- Lower resolution images reduce accuracy due to loss of important details.
- The system requires computational resources for training deep learning models.
- The approach is limited to image-based classification and does not support real-time deployment.
- The model may not generalize well to unseen defect types outside the dataset.

5.3 Future Enhancements

- Implement advanced techniques such as self-supervised learning to reduce dependency on labeled data.
- Incorporate attention mechanisms to improve defect detection accuracy.
- Optimize models to reduce computational cost and improve efficiency.
- Extend the system for real-time defect detection and deployment.
- Use larger and more diverse datasets to improve generalization.

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